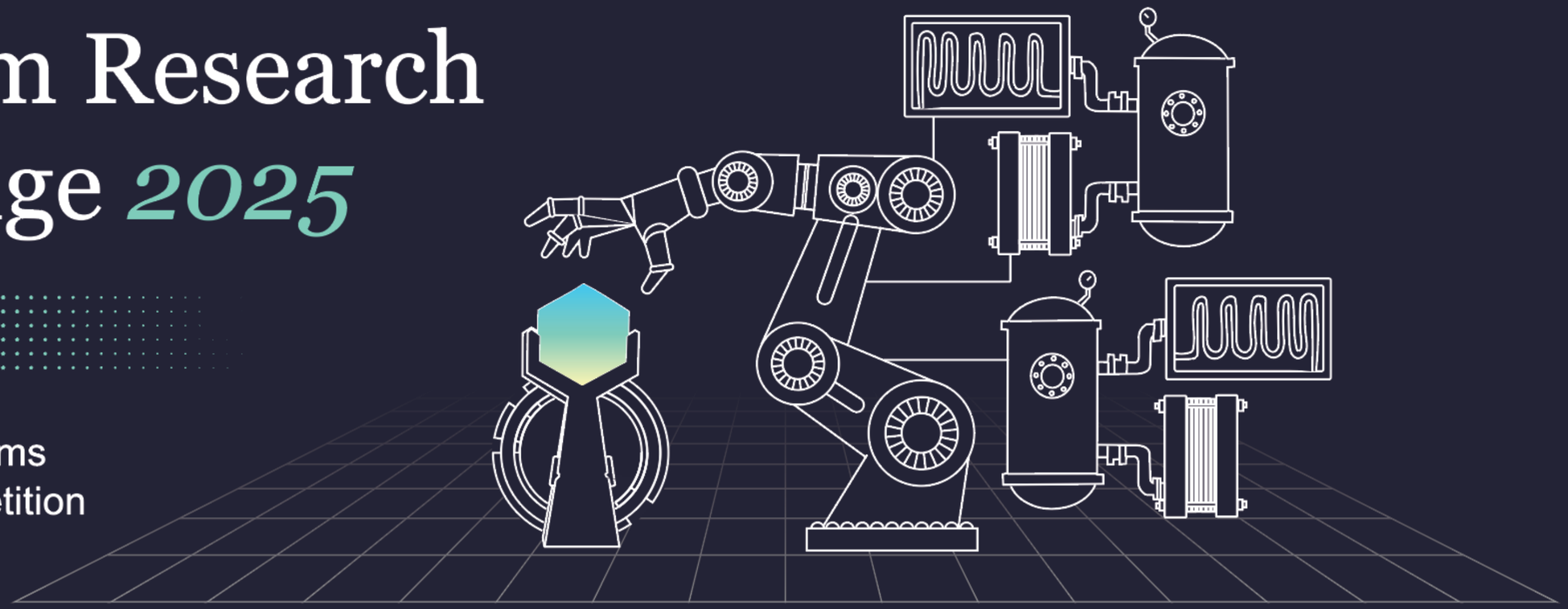
The Lam Research Challenge *2025*

A nationwide systems engineering competition



Grand Finale

Problem Statement

Problem Statment

Gates 1: Pump Switch ON

Gates 2: Peltier Switch ON

Gate 3: Lam LEDs Switch ON

Gate 4: HOLOGRAM ON

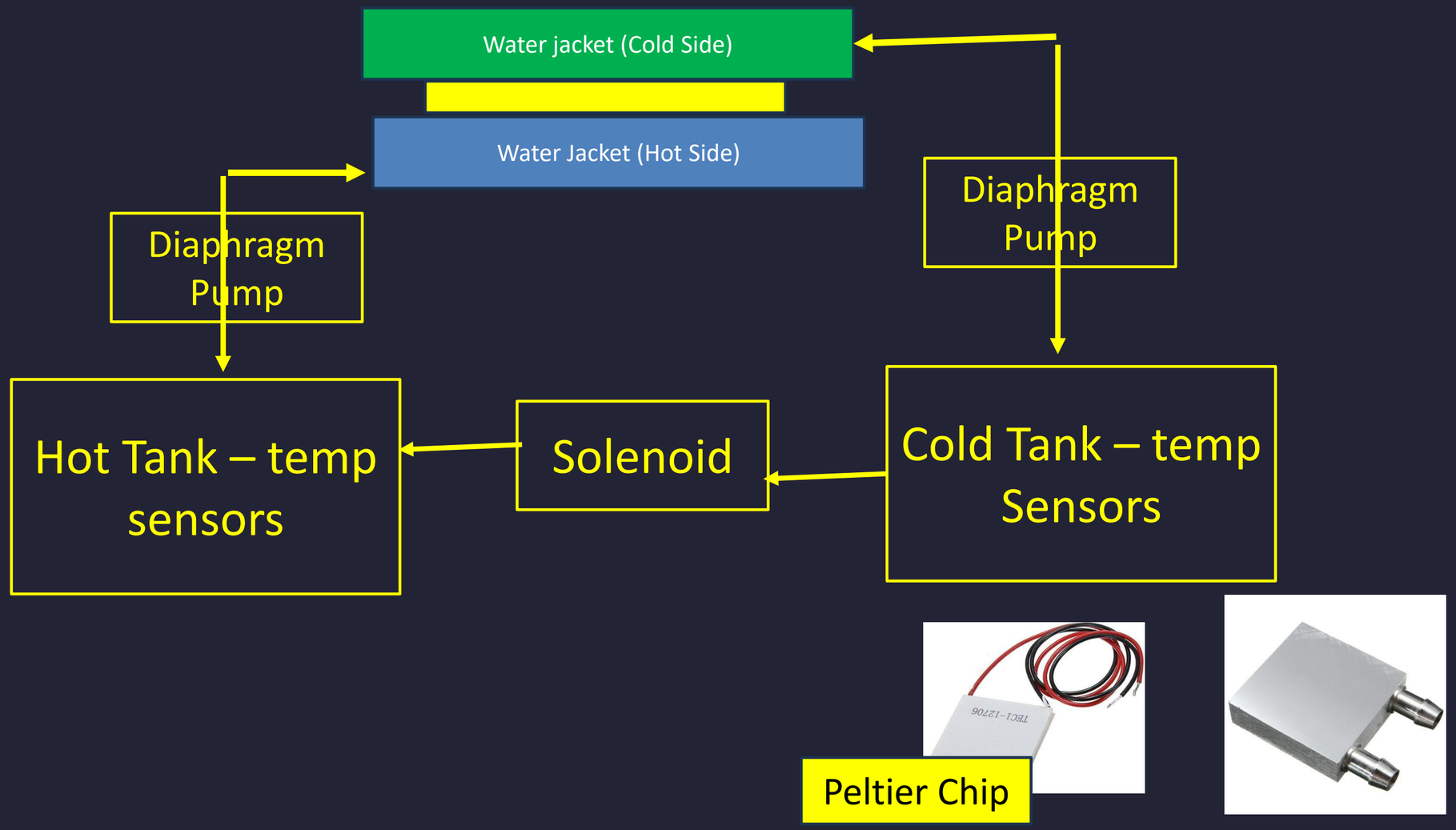
Temperature Monitoring & Controlling

A heat exchanger system needs to be designed in which water circulates between two tanks. The objective is to continuously monitor the temperature of both tanks and maintain a temperature difference (ΔT) of 20°C between them.

Components Given:

- Peltier module
- Water jacket
- Solenoid valve
- Two diaphragm pumps
- Pipes and valves

Project – Project Problem Statement



Gates 1&2 – Surprise Element in Arena

Heat Exchanger with Temperature Monitoring

Diaphragm Pump

Diaphragm Pump

Hot Tank – temp sensors

Cold Tank – temp Sensors

Thank you